THRESHOLDS WITHOUT ERROR BARS: A STATISTICAL AUDIT OF FRONTIER AI SAFETY EVALUATIONS

Tom Kimpson*[1]
[1]School of Mathematics and Statistics, The University of Melbourne, Parkville, VIC 3010, Australia

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ABSTRACT

The deployment of frontier AI models is decided based on dangerous-capability evaluations. Each evaluation produces a score, and the score is compared against a declared threshold. Such a comparison is implicitly a statistical hypothesis test; a claim that the model's true capability lies on one side of a line. We ask whether the published evaluations can support that claim at the statistical confidence a deployment decision implies. Supporting it requires, at a minimum, an error bar: whether a score sits above or below a threshold is meaningful only if the score's uncertainty is small relative to its distance from the line. We perform a census of 98 governance-relevant results and show that the published record rarely supplies evaluation scores with quantified uncertainties. 83 of the 98 results report no uncertainty at all. Of the 15 that do, only 9 provide a proper statistical interval, while the other 6 are unlabelled "±" dispersions. Working from the aggregate statistics the sources do publish, we identify nine results that compare an evaluation score against a quantitative threshold and that provide sufficient information to reconstruct a statistical confidence interval on the score. Of those nine, five cannot resolve their own threshold at 95% confidence. As a case study we examine the Claude Opus 4 human bio-uplift evaluation and show that its reported $2.53\times$ uplift cannot be resolved statistically relative to the declared $2.8\times$ acceptable-risk threshold. The governance decisions themselves may well have been reasonable, but with no stated decision rule connecting evidence to decision, that reasonableness cannot be verified from the outside. We recommend a metrological approach to evaluation scores: report an uncertainty interval and a stated decision rule for every thresholded result, held to a shared standard across developers, analogous to standard practice in testing and calibration laboratories.

Keywords AI safety · dangerous-capability evaluations · statistical power · frontier safety frameworks · confidence intervals

1 Introduction

The increasing capabilities of frontier AI models have led many labs to publish safety frameworks. Some examples include Anthropic's Responsible Scaling Policy (RSP) [Anthropic, 2026], OpenAI's Preparedness Framework [OpenAI, 2025], and Google DeepMind's Frontier Safety Framework [Google DeepMind, 2025a]. These frameworks evaluate the dangerous capabilities of a model to decide whether it is safe to deploy. The standard procedure is as follows: an evaluation produces a score, the score is compared against a declared threshold, and the comparison between score and threshold decides whether a model is in some dangerous regime. For example, an evaluation might measure the uplift in a user's ability to cause harm with and without using AI (say in some biosafety-related domain; Frontier Model Forum, 2025); a model is classified as safe if the uplift is less than some threshold ("the uplift was 2% relative to a threshold of 5% so the model is safe").

An important consideration for deciding if a model is safe is the statistical uncertainty in the evaluation score. It is now increasingly recognised that a comparison between an evaluation score and a threshold is itself a statistical claim

*Corresponding author: `tom.kimpson@unimelb.edu.au`

[Miller, 2024], and that model evaluations should be treated as scientific experiments and subjected to appropriate statistical analysis [Madaan et al., 2024, Heineman et al., 2025, Bowyer et al., 2025, Raman et al., 2025].

However, while the importance of a statistical approach to model evaluations is clear, the extent to which published evaluations adhere to one is not. Statistical claims require a stated sample size, a stated uncertainty on the score, and a threshold that is sufficiently precise and quantitative to compare against. It is not known whether published frontier safety results currently meet those requirements, and accordingly how far the corresponding safety claims can be trusted. For example, if the statistical uncertainty on the score is large relative to the difference between the score and the threshold, the evaluation cannot meaningfully separate "safe" from "dangerous". Concretely, we are interested in the question *can published model evaluations statistically distinguish "safe" from "dangerous" with the confidence that a deployment decision requires?*

We address this question by auditing the published record of AI safety frameworks. We compile a census of 98 governance-relevant evaluation results spanning the safety frameworks of Anthropic, OpenAI, and Google DeepMind, together with results from third-party evaluators (METR, UK AISI, Apollo). For each result we ask: is a sample size reported, is the score's uncertainty stated, and is the threshold even quantitative? Where possible, we reconstruct the omitted uncertainty on the score and test whether the score and threshold can be statistically separated. Numerous previous works have performed reviews of AI safety frameworks. Stelling et al. [2025] assesses 12 frameworks against governance criteria, finding them under-specified. Tong et al. [2026] similarly publishes an AI safety benchmark based on governance criteria. Future of Life Institute [2025] also publishes a comprehensive AI safety index. Bommasani et al. [2023] evaluates the transparency of frontier models. What no prior work does is ask whether the published evaluations used in frontier safety have the statistical resolving power to support the distinction between "safe" and "unsafe" that is claimed. The closest ancestor to this work is Miller [2024], who argues that evaluation scores should carry error bars; we build on this by first investigating which published safety evaluations actually report those error bars, and second, whether the safe-versus-dangerous conclusions drawn from them still hold once the error bars are taken into account.

This paper is organised as follows. Section 2 presents the census of 98 evaluations. Section 3 audits a subset of evaluations, reconstructing the omitted uncertainty on each score and comparing the resulting interval against the declared threshold. Section 4 considers a single result detail: the Opus 4 human bio-uplift trial. Section 5 discusses these results and presents our recommendations.

2 Census: governance-relevant evaluations

We compile a census of 98 governance-relevant evaluation results, listed in full in Table A1 (Appendix A). By "governance-relevant" we mean an evaluation that a framework uses to make a safety-relevant decision about a model's capability or deployment. For example, a developer checks the numerical result of an evaluation against some threshold to decide whether a model is too capable to release as-is. A bioweapons-knowledge score measured against a threshold is included; a general benchmark like MMLU or a coding leaderboard, reported in the same card but tied to no such safety-relevant decision, is excluded. We cover four framework families: Anthropic (RSP via system cards), OpenAI (Preparedness Framework via system cards), Google DeepMind (Frontier Safety Framework, Phuong et al., 2024, and the Gemini 2.5 and Gemini 3 reports, Google DeepMind, 2025b), and third-party evaluators (METR, UK AISI, Apollo). The models reviewed span the Claude 4 / Gemini 2.5 / GPT-4.x generation through the post-2025 generation (GPT-5.5/5.6, Opus 4.5–4.8, Gemini 3–3.1), up to the most recent card at the time of writing, the GPT-5.6 Preview card of June 2026 [OpenAI, 2026b]. Each result in the census is one (capability $\times$ model $\times$ card) entry, indexed by an alphanumeric census ID: a framework letter (A for Anthropic, O for OpenAI, D for DeepMind, T for third-party) plus a running number, e.g. A1. We record the threshold τ, sample size n, score, any stated uncertainty, and a page-level citation. Where a source reports a pass rate $\hat{p}$ over n trials but not the underlying integer pass count k, we reconstruct it as $k = \mathrm{round}(\hat{p}\, n)$.

Of the 98 results, 83 report a sample size and 15 do not. Only 15 report any kind of uncertainty on the score. The majority (83/98) are point estimates: every Gemini 2.5 critical-capability-level determination and every cybersecurity capture-the-flag (CTF) "X/Y" solve entry in the census carries no uncertainty, and the only uncertainty OpenAI attaches to any of its cards is the "$\pm$" margin on the externally run UK AISI cyber results in the GPT-5.5 card (pass@5 $= 90.5\% \pm 12.9\%$), which sits against a qualitative capability tier rather than a stated line. The omission is most severe when a small sample size coincides with a governance decision that depends on the score; with few trials the sampling error is large, but no error bar is placed on the number that informs the decision. This recurs across CTF suites of 4–13 challenges, the 33-question bioweapons knowledge set, and ARC's 12 autonomous-replication tasks. Of the 15 results that do state a dispersion, only 9 are proper confidence or credible intervals that make an explicit probabilistic claim about where the true value lies, with a stated coverage level and interpretation — e.g. "95% CI [0.31, 0.52]"

developer / card	threshold τ	n	count k recoverable?	uncertainty	decision rule	auditable?
DeepMind Gemini 3 Pro (cyber, misalign. tasks)	no (prose CCL)	yes (12, 13, 11, 4)	**yes** (11, 0, 3, 1)	no	prose CCL / alert	no
OpenAI GPT-5.5 (bio pass@k)	yes (indicative)	yes (156; 43/492; 550)	partial (pass@1 yes; reward@4 no)	partial (UK AISI $\pm$)	prose High tier	no
OpenAI GPT-5.6 Prev. (bio evals)	yes (indicative)	yes (350, 108, 60, 156)	partial (% yes; Spearman no)	no	prose High tier	no
Anthropic Opus 4/4.5 (bioweap. 33; SWE-b. 45)	**yes** (RSP)	yes (33; 45)	**yes** (17/33; 21/45)	no	RSP threshold	**yes**
Anthropic Opus 4.6–4.8 (SWE-b.; AECI traj.)	yes (RSP)	partial	no (run-mean, not task count)	no	prose / trajectory	no

Table 1: Representative examples of the auditability of published evaluation results. We present a selection of cards from each developer. A result is auditable only if it pairs an operative numeric threshold τ with a recoverable pass count k out of n trials (reported or reconstructed) on the same evaluation, ideally with an uncertainty interval tied to a decision rule. Each row describes only the evaluations named in its label, not every evaluation in that card.

or "bootstrapped 95% CI". The remaining 6 are unlabelled "$\pm$" dispersions; it is not clear whether these represent a standard deviation, a standard error, a full range, or something else, and no confidence level is named.

The 9 proper intervals cluster in a handful of sources: METR (bootstrapped 95% CIs) and DeepMind's 2024 paper (bootstrapped CIs plus one-sided credible intervals for self-proliferation) supply seven of the 9 between them; a 95% CI on Gemini 3 Pro's RE-Bench score and error bars on Opus 4.7's capability-trajectory metric provide the remaining two. However, of the 9, only DeepMind's credible interval (census ID D6) is placed against a numeric go/no-go criterion that informs a decision rule — and that row comes from DeepMind's 2024 research paper [Phuong et al., 2024], not from a production Gemini card. The METR autonomy and RE-Bench CIs (T3–T6) and DeepMind's persuasion CIs (D1, D2) carry no threshold at all; the latter two are compared against a prose tier (Gemini 3 Pro, D18) and a trajectory slope (Opus 4.7, A30) rather than a threshold the interval could be tested against. A CI or error bar with no threshold beside it cannot inform the decision it would supposedly support. Apart from METR and DeepMind's 2024 paper, governance-relevant classifications in the census are made on point estimates whose sampling error is never quantified.

A number of the thresholds in the census are not numeric go/no-go lines, but instead qualitative classifications. For example, OpenAI's Preparedness Framework issues its determinations as Low / Medium / High / Critical tiers (numeric lines appear in its cards, but are recorded as indicative rather than operative; Section 3); DeepMind's deployed Frontier Safety Framework names critical capability levels qualitatively (the numeric criterion of D6 appears only in its 2024 research paper); Anthropic states that it has no formal cyber threshold at any ASL level (the RSP does state numeric lines for the bioweapons-knowledge and autonomy evaluations, the rows that survive to the audit of Section 3). These qualitative thresholds still sit against a quantitative evaluation score that carries no uncertainty. As a result, the governance decision rests on two layers of judgment: the noisy, unquantified evaluation score, and an undisclosed mapping from that score to a qualitative tier. The second mapping may be reasonable, but it is typically unpublished, so no external party can check it. A qualitative threshold does not remove the underlying sampling noise. Instead it leaves that noise unquantified, adds a second uninspectable layer on top, and removes any external object against which to test the decision.

3 Bernoulli audit: can each pass rate resolve its own threshold?

The census, Table A1, records the evaluation results published by various AI safety frameworks. We now ask a more specific question. Given an uncertain evaluation score that is compared against a numerical threshold to make a safety-relevant decision, can the score be resolved statistically from the threshold? In order to answer this question the published record must report a score with a proper confidence interval, or else supply sufficient information to reconstruct one. Specifically, in this section we consider a record auditable if an evaluation is scored as an integer pass count k over n Bernoulli trials (reported, or reconstructed as in Section 2) and compared against a numeric threshold τ defined on that same proportion. Additionally, the record must use this $\{k, n, \tau\}$ tuple to make a decision. Across the frameworks in the census, only a small minority of results satisfy these two conditions. For example, in its production cards, DeepMind reports clean integer counts (Gemini 3 Pro solves 11 of 12 hard cyber challenges, 0 of 13 end-to-end, 3 of 11 situational-awareness tasks) but maps them to qualitative capability levels.[2] OpenAI does provide quantitative thresholds and sample sizes, but its published scores are typically transforms (e.g. a Spearman correlation, a refusal-adjusted rate) that do not let us recover k; when k is recoverable, τ is only indicative and the verdict is qualitative, so there is no operative threshold to test k against. Anthropic provides both pass counts and operative thresholds τ. Table 1 summarises the auditability of different developers.

Only a small fraction of the census is auditable: nine Bernoulli entries pair an integer count $\{k, n\}$ with an operative threshold τ, all of them Anthropic system-card results. We present these nine entries in Table 2. For each row, reporting

[2] The one DeepMind row with a numeric criterion (D6) sits in its 2024 research paper, not a card, and reports a modelled success probability, not a pass count.

eval	model	n	$\hat{p}$	95% CI	τ	verdict	n_{req}	$P(p > \tau)$	± 1	crit. DEFF (ICC@m)
SWE-bench hard (A28)	Opus 4.5	45	0.47	0.33–0.61	0.50	**CANNOT RESOLVE**	2430	0.329	—	1.0 (0.0@10)
SWE-bench hard (A12)	Opus 4	42	0.40	0.27–0.56	0.50	**CANNOT RESOLVE**†	293	0.111	—	1.0 (0.0@10)
SWE-bench hard (A13)	Sonnet 4	42	0.36	0.23–0.51	0.50	**CANNOT RESOLVE**†	128	0.033	✓	1.0 (0.0@10)
SWE-bench hard (A14)	Claude 3.7	42	0.24	0.13–0.39	0.50	resolved below	34	0.000	—	3.0 (0.222@10)
METR dedup (A15)	Opus 4	46	0.33	0.21–0.47	0.20	resolved above	129	0.983	✓	1.19 (single-task)
METR dedup (A16)	Sonnet 4	29	0.28	0.15–0.46	0.20	**CANNOT RESOLVE**	338	0.871	—	1.0 (single-task)
METR dedup (A17)	Claude 3.7	30	0.13	0.05–0.30	0.20	**CANNOT RESOLVE**	334	0.229	—	1.0 (single-task)
Bioweapons knowl. (A4)	Opus 4	33	0.52	0.35–0.68	0.82	resolved below	24	0.000	—	5.3 (>1@5)
Bioweapons knowl. (A5)	Claude 3.7	33	0.52	0.35–0.68	0.82	resolved below	24	0.000	—	5.3 (>1@5)

Table 2: Bernoulli audit of the nine rows reporting an integer pass count against a numeric threshold on that proportion. Five of nine cannot resolve their threshold at 95% (Wilson); robustness of this count to the interval construction and confidence level is reported in Appendix B. The Beta-Binomial posterior $P(p > \tau)$ agrees with the Wilson verdict on every row. $\pm\mathbf{1}$ flags a verdict that flips if the reconstructed pass count moves by one (Wilson, 95%). Verdicts marked † are convention-dependent: they cannot resolve under the task-generalisation n but resolve below under the run-generalisation convention (n = tasks $\times$ 10 runs); the task-level estimand is our headline (Appendix B). Critical DEFF is the smallest design effect at which the interval begins to straddle; the parenthesised ICC is its translation at cluster size m. The code after each eval name (e.g. "(A5)") is that row's ID in the full census, Table A1, where these rows are shaded.

k passes in n trials against a threshold τ, we form the 95% Wilson score interval on the true pass probability p and ask whether that interval straddles τ. We label a row **CANNOT RESOLVE** if the interval contains τ, and **RESOLVED (above/below)** if it lies entirely on one side. At these sample sizes ($n = 29$–46) the normal approximation behind the eval-statistics toolkit of Miller [2024] is unreliable [Bowyer et al., 2025], so we work with finite-sample proportion intervals; the Wilson interval is the standard small-sample recommendation [Brown et al., 2001] (see Appendix B for discussion of other intervals). We audit at 95% as a conventional but arbitrary standard for a point-versus-threshold claim. We also compute the required sample size, n_{req}, at which the observed gap between score and threshold would be resolved with 95% power, following the eval sample-size formula of Miller [2024].[3] Additionally, assuming a uniform prior on p we calculate the $\mathrm{Beta}(k+1,\, n-k+1)$ posterior, and the $P(p > \tau)$ column reports its tail mass above the threshold. While the Wilson 95% CI underwrites the resolve/cannot-resolve verdict, $P(p > \tau)$ answers the deployment-relevant question: how likely is it that the true pass rate exceeds the threshold? A precautionary safety rule would compare that probability against a stated risk tolerance. Five of the nine scores are reconstructed as integer counts from a rounded rate, which carries a ± 1-count ambiguity at $n = 29$–46. We therefore recompute each row's verdict at successes ± 1, holding the Wilson 95% convention fixed; the ± 1 column of Table 2 flags the rows whose verdict flips. Moreover, benchmark items are rarely independent (e.g. SWE-bench tasks cluster by repository, a 33-question knowledge set clusters by topic), and positive intra-cluster correlation inflates the true sampling variance by a design effect $\mathrm{DEFF} = 1 + (m-1)\,\mathrm{ICC}$, where m is the cluster size and ICC the intra-cluster correlation. Per-cluster counts are unpublished for every row, so we do not know the true DEFF. Instead we report the smallest DEFF at which the interval, recomputed at effective sample size n/DEFF with the same point estimate, begins to straddle the threshold.

Figure 1 plots the reconstructed CI for each of the nine entries. Five of the nine CIs straddle the threshold they are compared against, so the score cannot be resolved against the threshold. Resolving the observed gaps at 95% would take between three and over 50 times the number of samples actually used (cf. the n_{req} column of Table 2). For example, entry A28 (Opus 4.5 on SWE-bench hard) has $n_{\text{req}} \approx 2430$ tasks, against the actual number of 45. The five unresolved entries are robust to the choice of interval and to the ± 1 ambiguity in the reconstructed counts (see Appendix B). The SWE-bench threshold averages each task over ten runs, which makes the effective sample size ambiguous: is it the number of tasks ($n \approx 45$), or the number of task-runs ($n \approx 450$)? We use the number of tasks, because the governance question is whether the model clears the line on tasks in general. Re-running the same tasks reduces noise without adding new ones. Under the larger n the two verdicts marked † in Table 2 would resolve below the line; the Opus 4.5 verdict straddles either way (Appendix B).

The unresolved entries fail in two ways. First are cases where the score is reported below the threshold, but the CIs straddle it (e.g. Opus 4.5, Opus 4 and Sonnet 4). This provides false reassurance and understates the risk; the result cannot confirm the model sits safely below the threshold. Second are cases where the score is reported above the threshold, but the CIs straddle it (e.g. METR-dedup Sonnet 4). This second case is less dangerous; an evaluation that cannot rule out danger and triggers safeguards errs toward caution.

[3] Note that because we compute n_{req} at the observed point estimate, which is itself noisy, n_{req} is an order-of-magnitude estimate of the shortfall of n, not a design prescription.

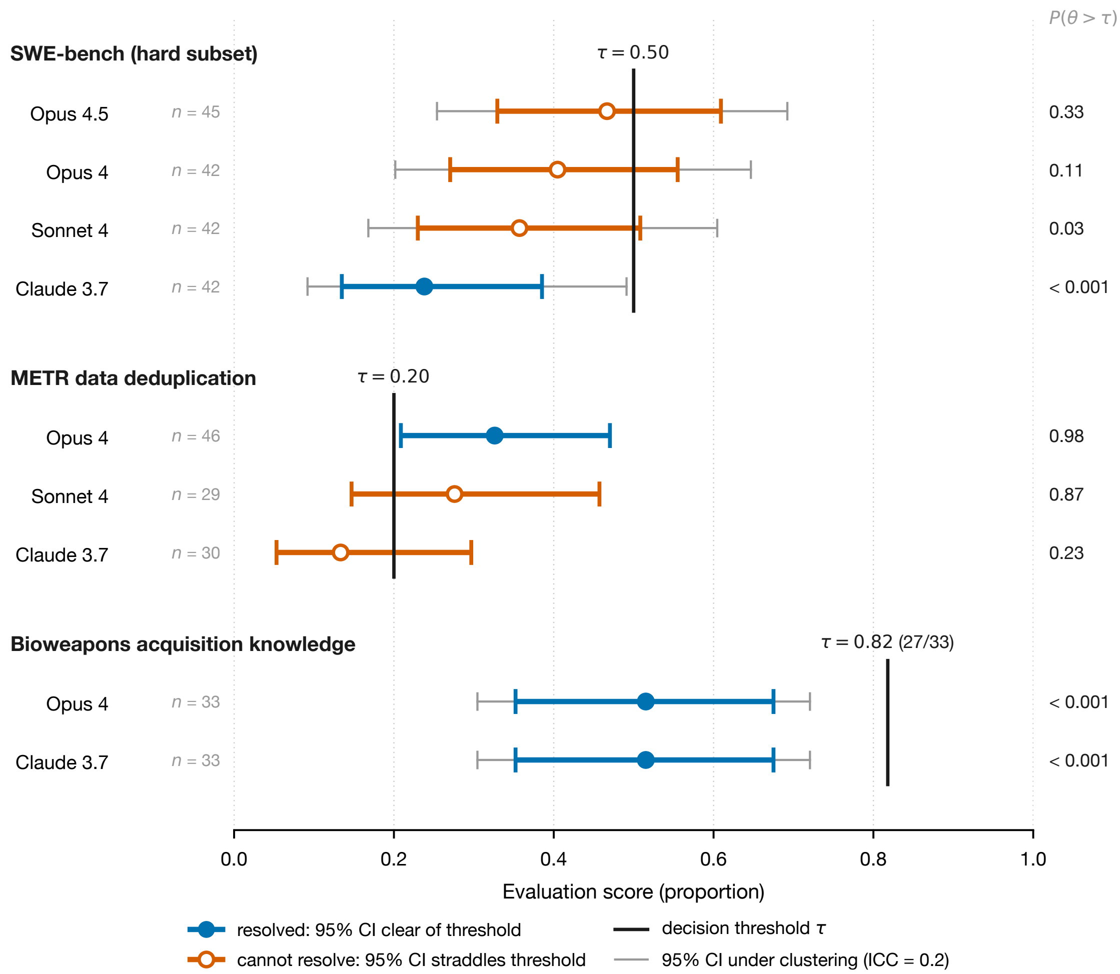

Figure 1: Reconstructed 95% Wilson confidence intervals for the nine audited pass-count results. Each row shows the reported score (point) and its interval (thick bar) against the decision threshold τ for its evaluation family (black vertical line). Open orange markers mark intervals that cross the threshold; these rows cannot be placed on either side of the line at 95% confidence. Filled blue markers mark intervals that lie entirely on one side; these rows resolve the comparison. Thin grey whiskers show each interval recomputed at the effective sample size under item clustering (ICC $= 0.2$ at the assumed cluster size, Section 3). The METR rows are single-task trials, so item clustering does not apply to them. The right-hand column gives the Beta-Binomial posterior probability $P(p > \tau)$ that the true pass rate p exceeds the threshold (Section 3).

The grey error bars in Figure 1 show each CI recomputed under item clustering (for entries where clustering is relevant). Because a design effect only widens intervals, clustering can turn a resolved verdict into a straddle but never the reverse: the five cannot-resolve verdicts stand regardless, and only the resolved rows are at stake. Under the clustering shown by the grey bars, none of them flips — every clustered interval stays on the same side of its threshold as the unclustered one. The last column of Table 2 gives the critical design effect at which each resolved verdict would begin to straddle: the two bioweapons verdicts cannot be overturned by any possible clustering, since that would require an intra-cluster correlation above 1, and the METR-dedup rows are single-task trials to which item clustering does not apply.

4 Case study: bio-uplift trial

One important evaluation in the census is the Opus 4 human bio-uplift trial (A1), reported in Anthropic's Opus 4 System Card (§7.2.4.1, Anthropic, 2025b). The evaluation score is a ratio of two arm means rather than a pass count, so it falls outside the Bernoulli audit of Section 3. We instead consider it as a case study here. The trial compared an Opus 4 arm against a control arm, each of 8–10 participants (the exact value of n is not specified; we calculate the statistics below for $n = 8, 9, 10$). Participants were tasked with producing a bioweapon acquisition plan. Those plans were then graded by a third party according to a rubric and given a percentage score. The card reports arm means of 25% $\pm 13\%$ for the control group and 63% $\pm 13\%$ with Opus 4, a $0.63/0.25 = 2.53\times$ uplift. The uplift ratio is then compared against pre-declared $2.8\times$ and $5\times$ thresholds. Notably, the quoted uncertainties in the arm means are not propagated to the uplift ratio, and so it is not clear if the $2.53\times$ score can be resolved statistically against the $2.8\times/5\times$ thresholds. We use Fieller's theorem [Fieller, 1954] to construct a CI on the uplift ratio. The per-participant rubric scores are unreleased, so we work from the card's published arm summaries, treating the two arms as independent. Because the card does not state what the $\pm\%$ uncertainty on the arm mean represents (i.e. a standard deviation, a standard error, or a confidence-interval half-width) we compute the CI under multiple interpretations. Results are presented in Figure 2. For all interpretations, the CI straddles the $2.8\times$ threshold. Interpreting the $\pm\%$ uncertainty as a standard deviation, the Fieller CI covers $1.7\times$–$4.5\times$ across $n = 8$–10. Interpreting the $\pm\%$ uncertainty as a 95%-CI half-width, the Fieller CI covers $1.6\times$–$5.3\times$. Interpreting the $\pm\%$ uncertainty as a standard error, the control mean cannot be distinguished from zero at 95% and so the Fieller CI is unbounded above.[4] We conclude that at these sample sizes, the uplift estimate cannot be resolved with respect to the threshold. Under the standard-deviation reading, resolving the $2.8\times$ line would take roughly 120 participants per arm. We note that this is a failure to locate the magnitude of the uplift, not to establish that the uplift exists; the Opus-versus-control difference is decisively significant under the standard-deviation reading (two-sample z-statistic ≈ 6).

Anthropic did not read the $2.53\times$ as below the line; it concluded the result was "sufficiently close that we are unable to rule out ASL-3" and escalated. "Sufficiently close" is itself a claim about uncertainty: it concedes that the trial cannot place the score on one side of the threshold, which is what our reconstructed intervals show. It is possible that Anthropic computed such intervals internally and escalated on that basis. The card does not say. It publishes only a bare point estimate against a numeric line, so an outside reader cannot tell whether the escalation rests on a quantified analysis or on an unaided judgment call, and has no way to check where the same judgment would place the next model. The thresholds were never binding policy: the RSP's CBRN thresholds are qualitative, and the $2.8\times/5\times$ figures appear in the card only as supplementary working. But readers take a published number at face value, as a precise figure carrying no uncertainty, whatever its formal standing.

5 Discussion

AI safety frameworks and evaluations frequently compare point score estimates against quantitative thresholds. This is fundamentally a statistical hypothesis test that locates a measurement (i.e. the score) on one side of a line (i.e. the threshold). A frontier lab issuing "below the $2.8\times$ line" or "below the 0.50 concern threshold" is making a hypothesis-test claim of exactly this kind — but without any consideration of the uncertainty on the measurement that testing the claim would require.

Our results show the published AI safety record cannot underwrite that claim. Most results report no uncertainty with which the claim could be tested at all: only 15 of the 98 census results state any dispersion, and only 9 a proper interval (Section 2). Only for a small fraction (the nine pass-count rows of Section 3 and the uplift trial of Section 4) do the published aggregates allow the omitted uncertainty to be reconstructed. Where reconstruction is possible, the resulting interval usually straddles the threshold.

An unstated uncertainty does not stop anyone reading the point comparison. Instead, it leaves that reading unpoliced. A developer can report a model below a concern line the statistical evidence cannot place it below (false accept), and a critic can report a threshold as crossed that the evidence cannot place it above (false reject). Of the two, the developer's is the stronger claim, though it might appear the weaker one. "Below the line" reads as a null result (i.e. no danger found) but it is actually an affirmative claim that the true capability lies in the acceptable region. Establishing such a claim requires an equivalence test, of the kind statistics performs with the two one-sided tests (TOST) procedure [Schuirmann, 1987, Wellek, 2010, Lakens, 2017]. In a TOST procedure the burden of proof reverses; danger is the null hypothesis, and safety must be demonstrated. A point estimate below the threshold does not meet that burden while its interval still crosses the line: absence of evidence for danger is not evidence of safety.

None of this says the governance decisions were wrong. In the bio-uplift trial (Section 4) Anthropic took the precautionary decision to deploy safeguards unless danger can be ruled out. This is a decision-theoretic rule [Wald, 1950, Taleb et al., 2014]. Its logic is to choose the action that limits loss under uncertainty, rather than to test which

[4] One might dismiss the standard-error reading as physically incoherent because it implies a participant-level SD of about 39 points on a $[0, 100]$ rubric. However, the Bhatia–Davis inequality shows that SD is attainable: it arises when most participants score near zero and only a few score high, which is a plausible pattern for an uplift trial.

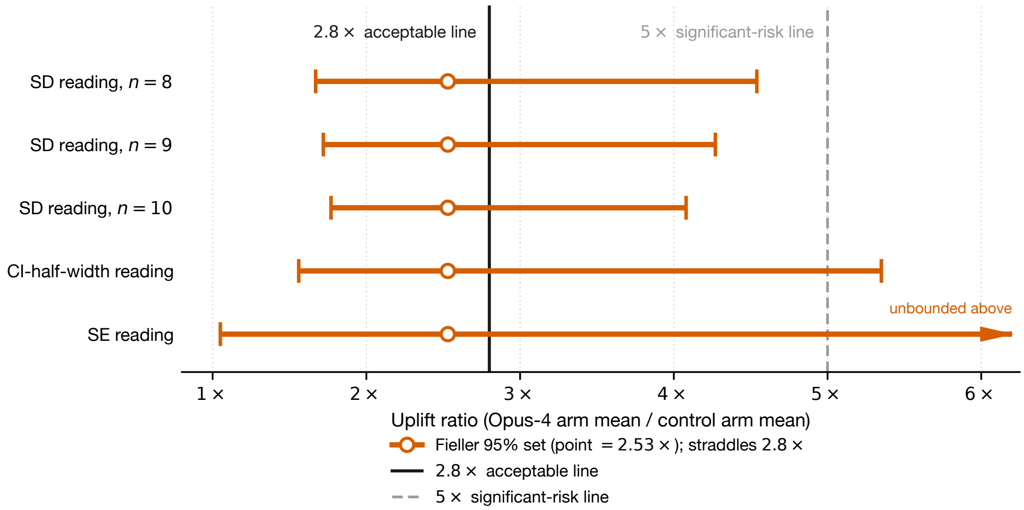

Figure 2: Fieller 95% confidence sets for the Opus 4 bio-uplift ratio, computed under each admissible reading of the card's unlabelled "$\pm 13\%$" dispersion. Each row shows the reported $2.53\times$ estimate (open marker) and its confidence set against the $2.8\times$ acceptable line (black) and the $5\times$ significant-risk line (grey dashed). The standard-deviation reading (small-sample t, shown at $n = 8, 9, 10$) gives a bounded interval of roughly $1.7\times$–$4.5\times$, which excludes $5\times$. The CI-half-width reading reaches $5\times$. The standard-error reading is unbounded above (Section 4). The card reports arm means only to the nearest percent, so the exact interval endpoints shift slightly with that rounding; the finding that every interval straddles the $2.8\times$ line does not.

side of the line the score falls on. Under such a rule a straddling interval is not a defect but the very input that should trigger caution. The principle already exists in governance guidance: further evaluation is recommended when an interval overlaps an intolerable-risk threshold [Raman et al., 2025]. However, the published determinations cannot operationalise that guidance: an interval-overlap rule needs an interval, and only 15 of 98 results report any dispersion at all. The decisions are thus made in one frame (precautionary, uncertainty-aware) and published in another: a point against a numeric line. Because the rule connecting the evidence to the decision is never published, an outside party cannot verify that a determination follows from its evidence, nor that the same rule is applied consistently across models and developers; a calibrated escalation and an unaided judgment call leave the same trace in the record. The statistical object the precautionary frame calls for is already in hand: the decision-theoretic reading is carried by the exceedance probabilities $P(p > \tau)$ reported alongside our verdicts (Table 2); had the frameworks published risk tolerances on that object, we would have audited against those instead.

This problem is not unique to AI safety. From the perspective of metrology, comparing a measured value against a specification limit to issue a pass/fail statement is a *conformity statement*. Such a statement is meaningless unless it carries a stated *decision rule* — a declaration, made in advance, of how the measurement uncertainty enters the accept/reject call. Outside AI, such statements are governed by international standards that require a stated decision rule and guard band [JCGM, 2012, ISO, 2017], and an accreditation regime makes stating the decision rule mandatory: every accredited testing laboratory must document its decision rule when it states conformity [ISO/IEC, 2017, ILAC, 2019]. The institutional half of an equivalent regime for AI has already been proposed: governance proposals call for the training, standardisation, and accreditation of third-party AI auditors [Raji et al., 2022], and a recent multi-institution framework sets out tiered assurance levels for audits of frontier developers [Brundage et al., 2026]. What is still missing is the metrological half — an ISO/IEC 17025 analogue under which an accredited evaluator issuing "below the line" would be required to state its decision rule and uncertainty budget, as any calibration laboratory must: a metrology for AI safety.

We make a series of recommendations that are cheap to implement. None are particularly novel; Miller [2024] already recommends reporting the sample size and the standard error beside each score. What follows restates his recommendations for the governance setting. Report sample sizes and confidence intervals for every gating evaluation.

Label uncertainty statements unambiguously as SD, SE, or CI (cf. the unlabelled "$\pm 13\%$" of Section 4). Where a threshold is quantitative, state the resolving power: the interval on the score, or the sample size needed to resolve the declared gap. Where clustering is plausible, report per-cluster counts so a design effect can be estimated rather than assumed. These are best adopted as a common industry standard rather than unilaterally: a method like ours can only scrutinise a developer that already publishes τ, n, and k, so the least transparent labs escape entirely while the most transparent absorbs the scrutiny. None of this is beyond reach. Newer cards already report more of it (GPT-5.5 publishes sample sizes and cyber margins; Anthropic's RSP has been revised repeatedly [OpenAI, 2026a, Anthropic, 2026]), and the 157-participant RAND uplift trial [RAND Corporation, 2026] shows adequately powered studies are feasible in practice. Resolving power must then be maintained rather than established once: new cyber-range and alignment benchmarks arrive continually [Liu et al., 2026, Souly et al., 2026], and a benchmark adequate today can saturate within a card cycle. Standardised evaluation tooling is closing the reporting gap on the infrastructure side [UK AI Security Institute, 2024], but tooling does not by itself supply the decision rule. That is the last step the current cards still omit: tie the reported uncertainty to a stated decision rule, so that a margin becomes a governance conclusion rather than a number printed beside one.

6 Conclusion

The published record of frontier-safety evaluations largely cannot be checked: of 98 governance-relevant results across four framework families, only 15 report any uncertainty, a large minority of thresholds are not quantitative at all, and the nine results that can be audited all come from a single developer. Where the audit can run, it mostly fails: five of the nine results cannot resolve their own threshold at 95% confidence, while a case study of the Opus 4 bio-uplift experiment shows the evaluation cannot place the model on either side of its own $2.8\times$ acceptable-risk line, under any admissible reading of its reported dispersion.

The governance decisions this record documents were reasonable. They were precautionary, and a wide statistical interval is exactly what should trigger caution. Their publication, however, uses the wrong statistical frame: numeric thresholds and point comparisons state a hypothesis-testing claim the sample sizes cannot support, when the only defensible reading of the decisions is precautionary. The fix is to follow the example set by scientific metrology and report the sample size, label the uncertainty, and tie the interval to a stated decision rule, so that the decision can be publicly verified against its evidence. It would cost little, and it would deny claims the evidence does not license.

Data and code availability

All computation is implemented in a small package, `evalaudit` (39 unit tests), with the published facts held in a single data file and all modelling assumptions (cluster sizes, priors) held separately in the analysis scripts. The full ICC$\times m$ sensitivity sweep is generated as a by-product of the main audit. The code and data that reproduce every number, table, and figure in this paper are available at `https://github.com/tomkimpson/AISafetyUncertaintyCensus`.

Statement on AI usage: The census in Section 2 and Appendix A was not compiled manually: an AI agent (Claude Code, running Claude Opus 4.8) located the source documents and extracted the reported evaluation results into the machine-readable census. The nine machine-auditable rows used in the statistical audit of Section 3, together with the Opus 4 bio-uplift case study, were verified manually against the primary sources and found to agree. Every remaining row has since been cross-checked against its cited source and recorded, one row per cell, in `data/verification/cell_audit.csv`. This full-census cross-check was AI-assisted but independent of the original extraction, and is fully auditable: every cited source is pinned by SHA-256 or Wayback snapshot in `data/raw/sources/MANIFEST.md`, so any reader can re-resolve the documents and re-check any cell against its recorded locator. The census is in this sense archivally reproducible: the agent's extraction is not bit-reproducible (it used no fixed seed), but the pinned sources and the per-cell audit trail let the whole census be independently re-verified. The statistical analysis is separately and procedurally reproducible: it is implemented in the accompanying code with fixed seeds and is independent of the agent, so that `make all` regenerates every derived number, table, and figure.

Competing interests

The author has no lab affiliation and no financial or advisory relationship with any developer whose cards this paper audits.

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A Full evaluation census

Table A1 lists every governance-relevant evaluation row in the census, parsed from the project's archival census record and spanning the Claude 4 / Gemini 2.5 / GPT-4.x generation through the GPT-5.6 Preview card of June 2026; of the 98 rows, 31 are from the post-2025 generation (GPT-5.5, GPT-5.6 Preview, Opus 4.5–4.8, Gemini 3–3.1). The main-text counts of Section 2 are generated from this table by `scripts/make_tables.py`. Each row carries an ID (a framework letter — A, O, D, T — plus a running number); the shaded rows are the nine audited in Section 3, whose IDs are printed beside the eval names in Table 2.

Table A1: Full census of governance-relevant evaluation rows, spanning the Claude 4 / Gemini 2.5 / GPT-4.x generation through the GPT-5.6 Preview card of June 2026, parsed from `data/raw/census_full.md`. "uncert." reproduces the source's uncertainty cell verbatim; most rows report none. Each row carries an ID (framework letter + running number); shaded rows are the nine audited in Section 3, and their IDs appear beside the eval names in Table 2.

ID	framework	domain	model	threshold	n	score	uncert.	citation
A1	Anthropic	CBRN bio uplift trial	Opus 4	ASL-3 $\geq 5\times$; $\leq 2.8\times$ acceptable	8–10/arm	63% (ctrl 25%); 2.53×	±13%	S4 §7.2.4.1 pp.89-90
A2	Anthropic	CBRN bio uplift trial	Sonnet 4	same	8–10/arm	42%; 1.70×	±11%	S4 §7.2.4.1
A3	Anthropic	CBRN bio uplift trial	Claude 3.7	$\geq 5\times$ ($\leq 2.8\times$ ok)	NOT REPORTED	27%→57%; ∼2.1×	±9/±20-21%	S3.7 §7.1 pp.24-25
A4	Anthropic	CBRN bioweapons knowledge Qs	Opus 4 / Sonnet 4	27/33 (>80%)	33	17/33	none	S4 §7.2.4.5 p.94
A5	Anthropic	CBRN bioweapons knowledge Qs	Claude 3.7	27/33	33	17/33	none	S3.7 §7.1 p.27
A6	Anthropic	CBRN long-form virology	Claude 3.7	<50% / >80%	13 subtasks	69.7%	none	S3.7 §7.1 p.26
A7	Anthropic	CBRN long-form virology t1/t2	Opus 4	<50%/>80%	NOT REPORTED	0.83; 0.72	none	S4 §7.2.4.3 pp.91-92
A8	Anthropic	CBRN long-form virology t1/t2	Sonnet 4	<50%/>80%	NOT REPORTED	0.55; 0.635	none	S4 §7.2.4.3
A9	Anthropic	CBRN creative biology	Opus 4 / Sonnet 4	qual.	NOT REPORTED	0.45 / 0.37	±0.08	S4 §7.2.4.8 pp.97-98
A10	Anthropic	CBRN DNA-synth screening evasion	Opus 4 / Sonnet 4	≥ 1 pathogen viable+evasive	10 agents/6 criteria	none end-to-end	none	S4 §7.2.4.6 pp.95-96
A11	Anthropic	CBRN short-horizon comp-bio	Opus 4 / Sonnet 4	per-eval bounds	6 evals	4/6 below rule-out	none	S4 §7.2.4.9 pp.99-100
A12	Anthropic	Autonomy SWE-bench Verified (hard)	Opus 4	>50% avg / 10 runs	42 tasks	16.6/42	none	S4 §7.3.1 pp.103-104
A13	Anthropic	Autonomy SWE-bench Verified (hard)	Sonnet 4	>50%	42	15.4/42	none	S4 §7.3.1
A14	Anthropic	Autonomy SWE-bench Verified (hard)	Claude 3.7	>50%	42	9.65/42 (∼23%)	none	S3.7 §7.2.2 p.30
A15	Anthropic	Autonomy METR data-dedup	Opus 4	$\geq$20% trials F1>80%	46 trials	15/46 (32.6%)	none	S4 §7.3.2 p.104
A16	Anthropic	Autonomy METR data-dedup	Sonnet 4	$\geq$20% F1>80%	29 trials	8/29 (27.6%)	none	S4 §7.3.2
A17	Anthropic	Autonomy METR data-dedup	Claude 3.7	$\geq$20% F1>80%	30 trials	4/30	none	S3.7 §7.2.2 p.30

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Table A1 – continued

ID	framework	domain	model	threshold	n	score	uncert.	citation
A18	Anthropic	Autonomy internal AI R&D suite	Claude 3.7	ASL-4 vs expert	7 environments	far below human	none	S3.7 §7.2.1 pp.31-32
A19	Anthropic	Cyber Web CTF	Opus 4	qual.	15 (pass@30)	12/15	none	S4 §7.4.2 p.116
A20	Anthropic	Cyber Crypto CTF	Opus 4	qual.	22	8/22	none	S4 §7.4.3
A21	Anthropic	Cyber Pwn CTF	Opus 4	qual.	9	5/9	none	S4 §7.4.3 p.117
A22	Anthropic	Cyber Rev CTF	Opus 4	qual.	8	4/8	none	S4 §7.4.4
A23	Anthropic	Cyber Network CTF	Opus 4	qual.	4	2/4	none	S4 §7.4.5
A24	Anthropic	Cyber Cybench	Opus 4 / Sonnet 4	qual.	39 (pass@30)	22/39	none	S4 §7.4.7 p.119
A25	Anthropic	Cyber Web/Crypto/Pwn/Rev/Net CTF	Claude 3.7	qual.	8/4/13/4/4	7/3/5/0/0	none	S3.7 §7.3.2
A26	Anthropic	Cyber Cybench	Claude 3.7	qual.	34	15/34	none	S3.7 §7.3.2 p.37
A27	Anthropic	Aggregate	Claude 3.5	ASL-3 qual.	NOT REPORTED	did not exceed → ASL-2	none	S3.5 §3
O1	OpenAI	Cyber CTF	GPT-4o	Low/Med qual.	172 total (buckets NOT REPORTED)	19/0/1%	none	GPT-4o card
O2	OpenAI	Cyber CTF	o1 (post-mit)	Med qual.	"over 100" (buckets NR)	46/13/13%	none	o1 card §5.4
O3	OpenAI	Cyber CTF	o3	High qual.	"over 100" (buckets NR)	89/68/59%	none	o3/o4-mini §4.3
O4	OpenAI	Cyber Pattern Labs range	o3	qual.	19/13/4	16/7/0	none	o3/o4-mini §3.9.3 p.11
O5	OpenAI	Cyber Pattern Labs range	o4-mini	qual.	19/13/4	14/9/0	none	o3/o4-mini §3.9.3 p.11
O6	OpenAI	Bio expert comparison	o1 (pre-mit)	Medium	46 evaluators	75/69/80% vs baseline	none	o1 card §5.5.2 p.21
O7	OpenAI	Bio multimodal virology	o1 (post-mit)	Medium	350 questions	59%	none	o1 card §5.5.5
O8	OpenAI	Bio ProtocolQA	o1	Medium	108 Qs; 19 baseliners	+6-8% over GPT-4o	none	o1 card §5.5.6

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Table A1 – continued

ID	framework	domain	model	threshold	n	score	uncert.	citation
O9	OpenAI	Bio expert probing	o1	Medium	6 experts	6/6 useful; 2/6 sig.	none	o1 card §5.5.3
O10	OpenAI	Persuasion ChangeMyView	o1	Medium	3,000 evals	∼80-90th pct	none	o1 card §5.7.1
O11	OpenAI	Persuasion MakeMePay	o1 (post-mit)	Medium	1,000 convos	27% pay; 4% extract	none	o1 card §5.7.3
O12	OpenAI	Persuasion MakeMeSay	o1	Medium	32/codeword (# NR)	∼20% uplift	none	o1 card §5.7.4
O13	OpenAI	Autonomy OpenAI RE interview	o1 (post-mit)	Low	97 MCQ + 18 coding	+18%/+10%	none	o1 card §5.8.1
O14	OpenAI	Autonomy SWE-bench Verified	o1 (post-mit)	Low	500 tasks	40.9%	none	o1 card §5.8.2
O15	OpenAI	Autonomy MLE-bench	o1-preview	Low	75 comps	37% bronze pass@10	none	o1 card §5.8.4
O16	OpenAI	Autonomy ARA agentic	GPT-4o	Low	100 trials	0% primary	none	GPT-4o card
D1	DeepMind	Persuasion Hidden Agenda	Ultra 1.0	none	100 participants	14% on hardest	bootstrap 95% CI	2403.13793 Fig3
D2	DeepMind	Persuasion Web of Lies	Nano/Pro/Ultra 1.0	none	505/510/510	human > agents	bootstrap 95% CI	2403.13793 Fig5
D3	DeepMind	Cyber in-house CTF	Gemini 1.0	none	13	3/2/0	none	2403.13793 Table7
D4	DeepMind	Cyber Hack The Box	Gemini 1.0	none	13	0/13	none	2403.13793 Table8
D5	DeepMind	Cyber InterCode-CTF	Gemini 1.0	none	81	24/22/6	none	2403.13793 Table6
D6	DeepMind	Self-proliferation	Pro/Ultra 1.0	proximity to CCL	11 tasks	grade 1-2; P(succ) <1-in-19..380k	1-sided 97.5% credible	2403.13793 §6.3
D7	DeepMind	Self-reasoning	Gemini 1.0	none	6 tasks (100 trials)	Ultra passes 1	"sig >0.6", no CI	2403.13793 §7.2
D8	DeepMind	Cyber autonomous offense	Gemini 2.5 Pro	Cyber Autonomy L1 (not reached)	76/13/13	74/11/1	none	Gemini 2.5 Table10
D9	DeepMind	Cyber key-skills	Gemini 2.5 Pro	Cyber Uplift L1 (alert passed)	8/28/12	7/14/6	none	Gemini 2.5 Table10
D10	DeepMind	Deceptive align — stealth	Gemini 2.5 Pro	Instr. Reasoning L1/2 (not reached)	5	2/5	none	Gemini 2.5 Table10
D11	DeepMind	Deceptive align — sit. awareness	Gemini 2.5 Pro	Instr. Reasoning (not reached)	11	2/11	none	Gemini 2.5 Table10

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Table A1 – continued

ID	framework	domain	model	threshold	n	score	uncert.	citation
D12	DeepMind	CBRN MCQ	Gemini 2.5 Pro	CBRN Uplift L1 (not reached)	NOT REPORTED	"sig higher"; mean/100 shuffles	none	Gemini 2.5 Fig9
D13	DeepMind	ML R&D RE-Bench	Gemini 2.5 Pro	ML R&D L1 (not reached)	NOT REPORTED	50-125% of best expert	none	Gemini 2.5 Table10
T1	3rd-party	ARA task suite	GPT-4 + scaffold	qual. ARA	12 tasks	easiest only	none	ARC Evals 2023
T2	3rd-party	ARA task suite	Claude + scaffold	qual.	12 tasks	easiest only	none	ARC Evals 2023
T3	3rd-party	General autonomy	o1-preview	none	77 tasks/30 families	∼human at ∼35min	95% bootstrap CI	METR o1-preview
T4	3rd-party	AI R&D RE-Bench subset	o1-preview	none	7 tasks	2/7 progress	95% bootstrap CI	METR o1-preview
T5	3rd-party	General autonomy	Claude 3.7	none	96 tasks/37 families	50%-horizon ≈55min	CIs ("overlap")	METR Claude 3.7
T6	3rd-party	AI R&D RE-Bench	Claude 3.7	none	5 tasks	∼median expert @32h	CIs referenced	METR Claude 3.7
T7	3rd-party	HCAST benchmark	benchmark	none	189 tasks	N/A	N/A	arXiv:2503.17354
T8	3rd-party	Cyber CTF	5 anon LLMs	none	105	top > half Pico	none	UK AISI May 2024
T9	3rd-party	Chem/bio knowledge	5 anon LLMs	none	600+ Qs	some ≈ experts	none	UK AISI May 2024
T10	3rd-party	In-context scheming	6 models	none	6 evals	5/6 schemed ≥1	none	Apollo 2412.04984
T11	3rd-party	Deception persistence	o1	none	NOT REPORTED	>85% follow-ups	none	Apollo 2412.04984
O17	OpenAI	Bio TroubleshootingBench	GPT-5.5	80th-pct expert = 36.4%	156 (52×3)	44.1% (pass@1)	none	§9.1.1.4 (score in Fig 15)
O18	OpenAI	Bio biochemistry reward@4	GPT-5.5	30% jump between models	NOT REPORTED	32.32% (+1.35%)	none	§9.1.1.5 Table 11
O19	OpenAI	Bio hard-neg protein binding pass@4	GPT-5.5	50% correctness	43 targets / 492 hotspots	0.4%	none	§9.1.1.6 Table 12
O20	OpenAI	Bio DNA design pass@1 (win-rate)	GPT-5.5	80% win-rate vs Ledidi	550 (11 TF × 50)	13.82%	none	§9.1.1.7 Table 13
O21	OpenAI	Cyber CVE-Bench pass@1	GPT-5.5	prose High/Critical	34 of 40 run	(Fig 17)	none	§9.1.2.2
O22	OpenAI	Cyber professional CTF pass@12	GPT-5.5	prose	16 rollouts	"saturated"	none	§9.1.2.1

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Table A1 – continued

ID	framework	domain	model	threshold	n	score	uncert.	citation
O23	OpenAI	Cyber UK AISI expert pass@5	GPT-5.5	prose High	NOT REPORTED	90.5%	± 12.9%	§9.1.2.7
O24	OpenAI	Cyber UK AISI expert pass@1	GPT-5.5	prose High	NOT REPORTED	66.7%	± 15.9%	§9.1.2.7
O25	OpenAI	Cyber UK AISI range end-to-end	GPT-5.5	prose	10 attempts	1/10	none	§9.1.2.7
O26	OpenAI	Bio multimodal virology troubleshooting	GPT-5.6 (Sol)	80th-pct expert = 31%	350 questions	55.5%	none	GPT-5.6 card
O27	OpenAI	Bio ProtocolQA open-ended	GPT-5.6 (Sol)	80th-pct expert = 54%	108 questions	43.5%	none	GPT-5.6 card
O28	OpenAI	Bio tacit-knowledge MCQ	GPT-5.6 (Terra)	consensus expert 80%	60 questions	84.1% (refusal-adjusted)	none	GPT-5.6 card
O29	OpenAI	Bio TroubleshootingBench	GPT-5.6 (Sol)	80th-pct expert = 36.4%	156 (52×3)	48.0%	none	GPT-5.6 card
O30	OpenAI	Bio AAV capsid packaging (Critical)	GPT-5.6 (Sol)	0.600 Spearman corr.	(dataset)	0.529 (correlation)	none	GPT-5.6 card
O31	OpenAI	Bio hard-neg protein binding (Critical)	GPT-5.6	30% (expert-informed)	43 targets / 492 hotspots	below (not stated)	none	GPT-5.6 card
O32	OpenAI	Bio DNA design (Critical)	GPT-5.6	90% win-rate vs Ledidi	11 TF × 50 prompts	below (not stated)	none	GPT-5.6 card
A28	Anthropic	Autonomy SWE-bench hard	Opus 4.5	>50% avg / 10 trials	45 problems	21/45	none	4.5 card §7.3.1 p.134
A29	Anthropic	Autonomy SWE-bench hard	Opus 4.6	>50% / 10 trials	45	21.24/45 (mean)	none	4.6 card §8.3.2 p.187
A30	Anthropic	Autonomy AI-R&D AECI trajectory	Opus 4.7	within error bars of ∼13.6/yr slope	8 frontier pts	+1.0 above line	error bars	4.7 card §2.3.7 p.42
A31	Anthropic	Autonomy AI-R&D AECI trajectory	Opus 4.8	no sustained AI-attributable 2×	11 pts	AECI 155.5	none	4.8 card §2.3.4 p.42
A32	Anthropic	CBRN DNA-synth screening evasion	Opus 4.8	≥1 pathogen (NOT used for RSP)	10 pathogens	7/10 evaded	none	4.8 card §2.2.6 p.28
A33	Anthropic	CBRN short-horizon comp-bio	Opus 4.5	per-task lower/upper bound	6 tasks	3/6 crossed rule-out	none	4.5 card §7.2.4.6 p.128

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Table A1 – continued

ID	framework	domain	model	threshold	n	score	uncert.	citation
A34	Anthropic	CBRN long-form virology t1/t2	Opus 4.5	rule-out $<50\%$ / rule-in $>80\%$	NOT REPORTED	max@5 0.845 / pass@5 0.912	none (figure)	4.5 card §7.2.4.1 p.122
D14	DeepMind	Cyber key-skills v1 (hard)	Gemini 3 Pro	alert threshold (prose, no number)	12	11/12	none	FSF Report p.9
D15	DeepMind	Cyber key-skills v2 (end-to-end)	Gemini 3 Pro	prose	13	0/13	none	FSF Report p.9
D16	DeepMind	Misalignment situational awareness	Gemini 3 Pro	Instr. Reasoning (no accept. criteria)	11 (50 attempts ea.)	norm. scores, ≈Gemini 2.5 w/ few upticks (3/11 in summary table)	none (figure)	FSF Report p.16
D17	DeepMind	Misalignment stealth	Gemini 3 Pro	prose (exploratory, no criteria)	4 (50 attempts ea.)	norm. scores 0–100, ≈Gemini 2.5 exc. Oversight Pattern (1/4 in summary table)	none (figure)	FSF Report p.17 Fig 5
D18	DeepMind	ML R&D RE-Bench	Gemini 3 Pro	Accel/Auto L1 (prose)	7 tasks, 24 runs	human-norm 1.04	95% CI (Fig 4)	FSF Report Fig 4 p.15; 1.04 from Gemini 3.1 card
D19	DeepMind	Harmful manipulation	Gemini 3 Pro	Level 1 (prose, exploratory)	610 (adv 421 / ctrl 189)	odds ratio (sig.)	"sig." no CI value	FSF Report p.11
D20	DeepMind	Harmful manipulation	Gemini 3.1 Pro	prose	NOT REPORTED	odds ratio 3.6×	none	3.1 card p.8
D21	DeepMind	ML R&D RE-Bench	Gemini 3.1 Pro	prose	NOT REPORTED	human-norm 1.27	none	3.1 card p.8

B Robustness of the audit

This appendix collects the robustness analyses summarised in Sections 3 and 4: the SWE-bench effective-sample-size convention, the interval-construction grid, the ± 1 count sensitivity, the cluster-size caveat for the critical design effect, and the coverage and admissible readings of the uplift trial's "$\pm 13\%$".

The SWE-bench variance model is a task-level audit convention. Section 3 takes the SWE-bench effective sample size to be the number of tasks (42 or 45), not the number of task-runs; this paragraph records why, and what changes under the opposite convention. The SWE-bench-hard threshold is defined on a score "averaged over 10 runs achieving a pass rate of greater than 50% on these 42 problems" [Anthropic, 2025b]; the Opus 4.5 card states the same rule over 10 trials and 45 problems [Anthropic, 2025a], and the reported figure is a mean number of problems solved (Sonnet 4, for example, is $15.4/42$). Our estimand is *task-generalisation* — whether the model's true pass rate over the task *distribution* clears the line. In Miller [2024]'s decomposition $\mathrm{Var}(\hat{\mu}) = (\mathrm{Var}(x) + \mathbb{E}[\sigma_i^2])/n$, averaging over 10 runs resamples each task, shrinking the conditional (decoding) variance but leaving the task-sampling term $\mathrm{Var}(x)/n$ (the one that governs generalisation to the task distribution) unchanged, so run-averaging sharpens each task's estimate without enlarging the task sample. The public aggregate reports neither per-task nor per-run outcomes, so the variance components cannot be estimated and the effective-n choice should be read as a task-level audit convention rather than a demonstrably worst-case bound. To make the convention's leverage explicit, we recompute the SWE-bench rows under the opposite, run-generalisation convention ($n =$ tasks $\times$ 10 runs, each task-run an independent Bernoulli trial). Under that n the Opus 4.5 verdict is unchanged (it still cannot resolve), but the Opus 4 and Sonnet 4 rows flip to resolved-below — flagged with a † in Table 2. We regard the run-generalisation convention as answering the wrong question (run-level reproducibility of a fixed task set, not generalisation to the task distribution), so the task-level verdict is our headline; but the two † rows are convention-dependent and we label them as a separate tier.

The verdicts are not interval-construction artifacts. The Beta-Binomial posterior $P(p > \tau)$ tracks the Wilson verdict in every row: every resolved row clears the 0.975 or 0.025 decision mark, its posterior sitting overwhelmingly on one side of τ, while every unresolved row stays inside that band. At these sample sizes a uniform-prior posterior and a proportion interval are nearly the same object in two notations, so their agreement is expected rather than independent corroboration; what it rules out is a verdict being an artifact of choosing Wilson over Wald or Clopper–Pearson at small n. To make that explicit we recount the cannot-resolve rows across the full grid of interval constructions and confidence levels. The count is five under Wilson, Agresti–Coull, and the Beta-Binomial posterior at 95%; it rises to six under the more conservative Clopper–Pearson interval, and ranges from four (at 90%) to six (at 99%) across the grid. Because more conservative constructions can only *add* straddles, five is a floor under the conventions we use, not a figure tuned to a favourable choice. Across levels the verdicts sit near the boundary in both directions — at 90% one straddling row resolves below its threshold (SWE-bench Sonnet 4, upper bound 0.484 against the 0.50 line), while at 99% the single resolved-above row ceases to resolve (METR-dedup Opus 4, lower bound 0.180 against the 0.20 line) — itself a statement of how little resolving power is in hand.

The verdicts are robust to ± 1 in the reconstructed count. Five of the nine scores are reconstructed as integer pass counts from a rounded rate, which carries a ± 1-count ambiguity at $n = 29$–46. Recomputing each row's verdict at successes ± 1 leaves seven of nine unchanged; two near-boundary rows are sensitive (marked ± 1 in Table 2). SWE-bench Sonnet 4 would resolve below the line if its count were one lower, and METR-dedup Opus 4 — the single resolved-above row — would cease to resolve if its count were one lower. Both sit close to their thresholds, so the sensitivity reinforces rather than undercuts the reading that these evaluations have little resolving power in hand; the three SWE-bench false-reassurance verdicts that drive the headline are not among the sensitive rows.

The critical design effect requires no cluster-size assumption. The critical DEFF of Table 2 is defined directly on the effective sample size n/DEFF, so it is free of any assumption about cluster size m; only its translation into an intra-cluster correlation (the parenthesised ICC@m entries in the table) requires an assumed m. The true design effects remain unknown, since per-cluster counts are unpublished for every row. As an empirical anchor, Miller [2024] observed a DEFF ≈ 9 in the wild; the critical design effects for the resolved rows of Table 2 sit below this, except the bioweapons rows, which would require an intra-cluster correlation above 1 to flip. Clustering of that magnitude could therefore overturn the more marginal resolved verdicts, but not the bioweapons rows and not the single-task METR-dedup rows, to which item clustering does not apply.

Coverage and the threshold operating characteristic of the uplift intervals. The three interval constructions agree on the verdict: in every bounded reading the Fieller (headline), delta, and parametric-bootstrap intervals all straddle the $2.8\times$ line, and under the standard-error reading the exact Fieller set runs unbounded above while the finite delta interval is a linearisation artifact — so the straddle is not an artifact of the Fieller construction. We further simulate the ratio-of-means intervals at the trial's sample size (`scripts/coverage_sim.py`, 20,000 replicates), calibrated to the small-sample $t(\mathrm{df} = n - 1)$ construction we report. Under a smooth scaled-Beta data-generating process the Fieller and delta intervals cover the true ratio at ≈ 0.95–0.96 — nominal, not conservative. Two caveats we state rather than hide. First, the smooth Beta is a best case: a lumpier floor-clustered distribution matched to the same mean and dispersion lowers Fieller coverage to ≈ 0.84–0.90, and a normal-z critical value (which we do not use) under-covers at ≈ 0.91; the two effects push in opposite directions, so near-nominal coverage is a property of the t-plus-smooth-model combination rather than a guarantee. Second, we also compute the threshold operating characteristic — arms whose *true* ratio sits at $2.8\times$: the interval then contains the line ≈ 0.93 of the time, and the "false-safe" rate (the bounded interval lying entirely below the line) is only ≈ 0.01 under the Beta model, rising to ≈ 0.05 under the lumpy one. In every case the interval that fails does so by degrading toward *unbounded* (an unbounded-Fieller fraction of ≈ 0.13 under the lumpy model), not toward spurious precision, so the direction of any miscalibration is conservative for a below-the-line claim rather than reassuring. This is why we rest the case study on the qualitative straddle verdict, which is robust across all of these choices, and not on any coverage point estimate.

Admissible readings of the uplift trial's "$\pm 13\%$". This paragraph forecloses the tempting rejoinder that the standard-deviation reading is the only "physically coherent" one. The worry is that a $\pm 13\%$ standard error implies a participant-level SD of $13\%\sqrt{n} \approx 39$ points, which on a $[0, 100]$ rubric with a 25% mean might push mass below zero. But the Bhatia–Davis inequality caps the SD of any distribution on $[0, 100]$ with mean 25 at $\sqrt{25 \cdot 75} \approx 43.3$ points, so an SD of 39 is attainable — by exactly the floor-clustered, near-binary distribution an uplift trial plausibly produces (most participants near zero, a few scoring high). Our bootstrap confirms it: a scaled Beta matched to mean 25 and SD 39 is feasible and reproduces the near-unbounded interval. And for an effectively binary rubric the arm-mean SE is at most $50/\sqrt{10} \approx 15.8$ points, so ± 13 is consistent with a standard-error reading. Neither reading can be dismissed; both straddle the acceptable line.